POLITECNICO DI MILANO

PhD in INGEGNERIA DELL'INFORMAZIONE / INFORMATION TECHNOLOGY - 42nd cycle

Research Area n. 1 - Computer Science and Engineering

THEMATIC Research Field: AUDITING AND MITIGATING EPISTEMIC FAILURES IN WEB-CONNECTED AI AGENTS

Monthly net income of PhDscholarship (max 36 months)
1500.0 In case of a change of the welfare rates during the three-year period, the amount could be modified.

Context of the research activity	
Motivation and objectives of the research in this field	AI agents increasingly mediate how people access online information by retrieving sources, synthesising evidence, personalising responses, and acting on behalf of users. Current evaluations mainly reward factual accuracy or task completion and therefore overlook broader epistemic failures. An agent may present weak or manipulated evidence with credible framing, obscure the provenance of information, amplify biased source distributions, adapt excessively to a user's beliefs through sycophancy, or reinforce errors over repeated interactions. These risks are especially significant in web-connected, multi-turn agents, where retrieval, memory, reasoning, and personalisation jointly shape what users know, trust, and believe. This PhD project will address the epistemic reliability of web-connected AI agents during information retrieval and synthesis. The research will develop measurable definitions, diagnostic methods, and practical safeguards for failures affecting source selection, evidence attribution, and claim formation. The main objectives are: 1. **Operationalise key epistemic failure modes**, including evidence laundering, source omission, overconfident synthesis, bias amplification,

	sycophancy, and susceptibility to adversarial or manipulated online content. 2. **Design reproducible benchmarks and longitudinal audits** to measure how these failures vary across models, tasks, interaction histories, degrees of personalisation, and information environments. 3. **Identify the mechanisms that cause or amplify epistemic failures** in retrieval, ranking, synthesis, memory, and reasoning through controlled experiments with open models and modular agent architectures. 4. **Develop and validate lightweight safeguards**, including source verification, provenance-aware generation, weak-model oversight, model steering, and interaction-design interventions, while preserving usefulness, personalisation, and user agency. Expected outcomes include new metrics and benchmark resources, empirical analyses of failure mechanisms, open-source evaluation tools, and design principles for more epistemically reliable AI agents.
Methods and techniques that will be developed and used to carry out the research	The research will combine **black-box auditing, controlled experiments, and focused user studies**. A benchmark suite will be developed to evaluate epistemic failures in web-connected agents, including evidence laundering, source omission, bias, sycophancy, overconfidence, and vulnerability to manipulated content. Longitudinal audits of frontier systems will be complemented by experiments with open models and modular retrieval pipelines. By varying retrieval, ranking, memory, prompting, and synthesis components, the research will identify where failures emerge and how they propagate across multi-turn and agent-agent interactions. Lightweight safeguards—such as source verification, provenance scoring, claim-evidence consistency checks, model-based oversight, steering, and uncertainty-aware interfaces—will then be tested. Their effectiveness will be evaluated against robustness, computational cost, usability, and preservation of usefulness and user agency.

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Educational objectives	The PhD candidate will: 1. Gain advanced theoretical and practical knowledge of large-language-model-based agents, retrieval-augmented generation, AI evaluation, and trustworthy AI. 2. Learn to design benchmarks, longitudinal audits, controlled experiments, and diagnostic analyses for complex AI and socio-technical systems. 3. Acquire hands-on experience with frontier and open-weight models, agent frameworks, retrieval systems, evaluation pipelines, model instrumentation, and human-subject studies. 4. Develop an interdisciplinary perspective integrating computer science, computational social science, information integrity, and human-computer interaction. 5. Strengthen skills in research ethics, reproducibility, scientific communication, international collaboration, and the dissemination of results through high-impact publications, conferences, workshops, and research projects.
Job opportunities	Potential career paths include academic positions and public research institutions, industrial research and development laboratories, technology companies developing or deploying AI agents, platform integrity and trust-and-safety teams, public-interest technology organisations, regulatory and standardisation bodies, and specialist AI governance or technology consultancies. The combination of technical auditing, experimental research, AI-system development, and human-centred evaluation will provide the candidate with a profile suitable for organisations working on the reliable

	deployment of AI in high-impact information environments.
Composition of the research group	1 Full Professors 1 Associated Professors 2 Assistant Professors 5 PhD Students
Name of the research directors	Prof. Francesco Pierri, Prof. Marco Brambilla

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Additional support - Financial aid per PhD student per year (gross amount)	
Housing - Foreign Students	--
Housing - Out-of-town residents	--

Scholarship Increase for a period abroad	
Amount monthly	750.0 €
By number of months	6

Additional information: educational activity, teaching assistantship, computer availability, desk availability, any other information
EDUCATIONAL ACTIVITIES (purchase of study books and material, including computers, funding for participation in courses, summer schools, workshops and conferences): financial aid per PhD student. TEACHING ASSISTANTSHIP: availability of funding in recognition of supporting teaching activities by the PhD student. There are various forms of financial aid for activities of support to the teaching practice. The PhD student is encouraged to take part in these activities, within the limits allowed by the regulations. COMPUTER AVAILABILITY: 1st year: Yes 2nd year: Yes 3rd year: Yes